

Safety of Toys

Part 1: Mechanical and Physical Properties, Part 2: Flammability

IS 9873 : 2017

1 Scope

1.1 This standard specifies requirements and test methods for toys intended for children up to 14 years of age. The purpose is to protect children from hazards that may arise from toys during use, foreseeable misuse, or ingestion of parts.

1.2 This standard covers mechanical and physical properties, flammability characteristics, chemical requirements, electrical properties, and hygiene requirements.

NOTE The following products are excluded from scope: sports equipment (skates, skateboards), bicycles, video game consoles, and collectible scale models.

2 Definitions

2.1 Toy - Any product or material designed or clearly intended for use in play by children under 14 years of age.

2.2 Small Part - Any part or component of a toy that is completely enclosed in the small parts cylinder as specified in Clause 5.2 when tested according to the method therein.

2.3 Small Part Cylinder - A cylinder 57.1 mm long and 31.7 mm in internal diameter, open at one end, with the closed end formed by a cylinder inclined at 45 degrees.

2.4 Accessible - Able to be contacted by a test probe as specified in Annex A of this standard.

3 Age Grading

3.1 The manufacturer shall specify the intended age range for each toy. Age grading shall be based on developmental stage, including cognitive, physical and emotional development.

3.2 Toys intended for children under 36 months (3 years) shall be labelled accordingly and shall comply with the enhanced requirements specified in Clause 4.3.

4 Mechanical and Physical Requirements

4.1 General

4.1.1 Toys and their components shall withstand the mechanical and physical tests specified in this clause without giving rise to hazards. Hazards to be prevented include sharp edges and sharp points, small parts that could be swallowed, strangulation risks from long cords and loops, and projectile

hazards.

4.2 Sharp Edges and Points

4.2.1 After the drop test and torque test, toys shall not have accessible edges that present a cutting hazard when assessed by the edge tester defined in IS 9873 (Part 1).

4.2.2 Rigid wire or rod-type accessible points shall not penetrate to the bottom of the Point Test Gauge.

4.2.3 The edge tester shall apply a force of 1 N +/- 0.1 N in the direction perpendicular to the edge.

Table 1 Mechanical Test Summary - Applicable Age Groups (IS 9873:2017)

Test	Method Clause	< 18 months	18-36 months	3-6 years	7-14 years
Drop Test	5.3	Yes	Yes	Yes	Optional
Torque Test	5.4	Yes	Yes	Yes	No
Tension/Compression	5.5	Yes	Yes	No	No
Sharp Edge Check	5.6	Yes	Yes	Yes	Yes
Sharp Point Check	5.7	Yes	Yes	Yes	Yes
Small Parts Check	5.8	Yes	Yes	No	No
Bite Test	5.9	Yes	No	No	No

4.3 Requirements for Toys for Children Under 36 Months

4.3.1 Small parts - Toys intended for children under 36 months shall not include small parts that fit entirely within the small parts cylinder.

4.3.2 Squeeze toys - Squeeze toys that can be mouthed shall not present a suffocation risk. The minimum opening of any aperture after compression shall not exceed 10 mm.

4.3.3 Long flexible parts - Cords, strings, and elastic shall not exceed 300 mm in free length for toys for children under 36 months.

5 Flammability

5.1 General Requirements

5.1.1 Toys shall not constitute a fire or burn hazard by themselves or when exposed to other ignition sources normally present in children play environments.

5.1.2 Materials used in the construction of toys shall comply with at least one of the categories specified in Table 2.

Table 2 Flammability Categories and Burning Rate Limits (IS 9873:2017)

Category	Material Type	Max Burning Rate (mm/s)	Test Method
F1	Pile fabrics and napped textiles	30	IS 9873 (Pt 2) Cl. 6.1
F2	Solid non-metallic materials	50	IS 9873 (Pt 2) Cl. 6.2
F3	Hair, beard, moustache materials	30	IS 9873 (Pt 2) Cl. 6.3
F4	Thin flexible non-metallic sheet	50	IS 9873 (Pt 2) Cl. 6.4

Category	Material Type	Max Burning Rate (mm/s)	Test Method
F5	Costumes and fancy dress outfits	30	IS 9873 (Pt 2) Cl. 6.5

5.2 Specific Prohibitions

- 5.2.1 Cellulose nitrate shall not be used in the construction of toys.
- 5.2.2 Toys shall not contain flammable liquid propellants.

6 Chemical Requirements

6.1 Migration Limits for Accessible Substrate Materials

6.1.1 The limits specified in Table 3 shall not be exceeded when tested per IS 9873 (Part 3).

Table 3 Element Migration Limits - Accessible Toy Materials (mg/kg) (IS 9873:2017)

Element	Scraped-off Material	Dry / Brittle Material	Liquid / Sticky Material
Aluminium (Al)	5625	1406	70000
Antimony (Sb)	45	11.3	560
Arsenic (As)	3.8	0.9	47
Barium (Ba)	4500	1125	56000
Boron (B)	1200	300	15000
Cadmium (Cd)	1.9	0.5	23
Chromium III	37.5	9.4	460
Chromium VI	0.02	0.005	0.2
Cobalt (Co)	10.5	2.6	130
Lead (Pb)	2.0	0.5	23
Mercury (Hg)	7.5	1.9	94
Nickel (Ni)	75	18.8	930
Selenium (Se)	37.5	9.4	460
Strontium (Sr)	4500	1125	56000
Tin (Sn)	15000	3750	180000